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LIGHT PATH FOLDING MODULE, CAMERA MODULE AND ELECTRONIC DEVICE

Abstract

A light path folding module includes a light path folding element and an opaque body. The light path folding element is for folding an incident optical axis to an exiting optical axis, and the light path folding element includes a light incident surface and a light exiting surface. The opaque body is disposed relative to the light path folding element, and the opaque body includes a non-closed ring structure and a plurality of light blocking structures. The exiting optical axis passes through a notch opening of the non-closed ring structure. The light blocking structures extend toward the exiting optical axis from the non-closed ring structure along a direction perpendicular to the exiting optical axis, wherein the light blocking structures are disposed adjacent to the light exiting surface of the light path folding element.

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Background/Summary

RELATED APPLICATIONS

[0001] This application claims priority to Taiwan Application Serial Number 113108658, filed Mar. 8, 2024, which is herein incorporated by reference.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a light path folding module and a camera module. More particularly, the present disclosure relates to a light path folding module and a camera module which are applicable to portable electronic device.

Description of Related Art

[0003] In recent years, portable electronic devices have developed rapidly. For example, intelligent electronic devices and tablets have been filled in the lives of modern people, and camera modules mounted on portable electronic devices have also prospered. However, as the technology advances, the quality requirements of camera module are becoming higher and higher. Therefore, developing a camera module that can improve the imaging quality has become an important and urgent problem in the industry.

SUMMARY

[0004] According to one aspect of the present disclosure, a light path folding module includes a light path folding element and an opaque body. The light path folding element is for folding an incident optical axis to an exiting optical axis, and the light path folding element includes a light incident surface and a light exiting surface. The incident optical axis passes through and enters into the light incident surface. The exiting optical axis passes through and is away from the light exiting surface. The opaque body is disposed relative to the light path folding element, and the opaque body includes a non-closed ring structure and a plurality of light blocking structures. The exiting optical axis passes through a notch opening of the non-closed ring structure. The plurality of light blocking structures extend towards the exiting optical axis from the non-closed ring structure along a direction perpendicular to the exiting optical axis, wherein the plurality of light blocking structures are disposed adjacent to the light exiting surface of the light path folding element. When the exiting optical axis as a center of circle is taken, an angle occupied by the plurality of light blocking structures is θ_e , and a number of the plurality of light blocking structures is N_e , the following conditions are satisfied: $10 \text{ degrees} < \theta_e < 350 \text{ degrees}$; and $15 < N_e < 460$.

[0005] According to another aspect of the present disclosure, a camera module includes the light path folding module according to the foregoing aspect and an imaging lens assembly. The imaging lens assembly is disposed adjacent to the light path folding module.

[0006] According to another aspect of the present disclosure, an electronic device includes the camera module according to the foregoing aspect and an image sensor. The image sensor is disposed on an imaging surface of the camera module.

[0007] According to another aspect of the present disclosure, a light path folding module includes a light path folding element and an opaque body. The light path folding element is for folding an incident optical axis to an exiting optical axis, and the light path folding element includes a light incident surface and a light exiting surface. The incident optical axis passes through and enters into the light incident surface. The exiting optical axis passes through and is away from the light exiting surface. The opaque body is disposed relative to the light path folding element, and the opaque body includes a non-closed ring structure and a plurality of light blocking structures. The incident optical axis passes through a notch opening of the non-closed ring structure. The plurality of light

blocking structures extend towards the incident optical axis from the non-closed ring structure along a direction perpendicular to the incident optical axis, wherein the plurality of light blocking structures are disposed adjacent to the light incident surface of the light path folding element. When the incident optical axis as a center of circle is taken, an angle occupied by the plurality of light blocking structures is θ_i , and a number of the plurality of light blocking structures is N_i , the following conditions are satisfied: $10\text{ degrees} < \theta_i < 350\text{ degrees}$; and $15 < N_i < 460$.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The present disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

[0009] FIG. 1A is a three-dimensional schematic view of a camera module of the 1st Example according to the 1st Embodiment of the present disclosure.

[0010] FIG. 1B is an exploded view of the camera module of the 1st Example according to the 1st Embodiment in FIG. 1A.

[0011] FIG. 1C is a cross-sectional view of partial of the camera module of the 1st Example according to the 1st Embodiment in FIG. 1A.

[0012] FIG. 1D is a plane view of the camera module of the 1st Example according to the 1st Embodiment in FIG. 1A.

[0013] FIG. 1E is a cross-sectional view along line 1E-1E in FIG. 1D.

[0014] FIG. 1F is a plane view of the opaque body of the 1st Example according to the 1st Embodiment in FIG. 1A.

[0015] FIG. 1G is a plane view of the opaque body of the 2nd Example according to the 1st Embodiment in FIG. 1A.

[0016] FIG. 2A is a three-dimensional view of a camera module of the 1st Example according to the 2nd Embodiment of the present disclosure.

[0017] FIG. 2B is an exploded view of the camera module of the 1st Example according to the 2nd Embodiment in FIG. 2A.

[0018] FIG. 2C is a plane view of the camera module of the 1st Example according to the 2nd Embodiment in FIG. 2A.

[0019] FIG. 2D is a cross-sectional view along line 2D-2D in FIG. 2C.

[0020] FIG. 2E is a plane view of the opaque body of the 1st Example according to the 2nd Embodiment in FIG. 2A.

[0021] FIG. 2F is a plane view of the opaque body of the 2nd Example according to the 2nd Embodiment in FIG. 2A.

[0022] FIG. 3A is a schematic view of a light path folding module of the 1st Example according to the 3rd Embodiment of the present disclosure.

[0023] FIG. 3B is an exploded view of the light path folding module of the 1st Example according to the 3rd Embodiment in FIG. 3A.

[0024] FIG. 3C is a plane view of the light path folding module of the 1st Example according to the 3rd Embodiment in FIG. 3A.

[0025] FIG. 3D is a schematic view of the opaque body of the 1st Example according to the 3rd Embodiment in FIG. 3A.

[0026] FIG. 3E is another plane view of the light path folding module of the 1st Example according to the 3rd Embodiment in FIG. 3A.

[0027] FIG. 3F is a cross-sectional view along line 3F-3F in FIG. 3E.

[0028] FIG. 4A is a schematic view of a light path folding module of the 1st Example according to the 4th Embodiment of the present disclosure.

[0029] FIG. 4B is an exploded view of the light path folding module of the 1st Example according to the 4th Embodiment in FIG. 4A.

[0030] FIG. 4C is a plane view of the light path folding module of the 1st Example according to the 4th Embodiment in FIG. 4A.

[0031] FIG. 4D is a schematic view of the opaque body of the 1st Example according to the 4th Embodiment in FIG. 4A.

[0032] FIG. 4E is another plane view of the light path folding module of the 1st Example according to the 4th Embodiment in FIG. 4A.

[0033] FIG. 4F is a cross-sectional view along line 4F-4F in FIG. 4E.

[0034] FIG. 5A is a schematic view of a light path folding module of the 1st Example according to the 5th Embodiment of the present disclosure.

[0035] FIG. 5B is an exploded view of the light path folding module of the 1st Example according to the 5th Embodiment in FIG. 5A.

[0036] FIG. 5C is a plane view of the light path folding module of the 1st Example according to the 5th Embodiment in FIG. 5A.

[0037] FIG. 5D is a schematic view of the opaque body of the 1st Example according to the 5th Embodiment in FIG. 5A.

[0038] FIG. 5E is another plane view of the light path folding module of the 1st Example according to the 5th Embodiment in FIG. 5A.

[0039] FIG. 5F is a cross-sectional view along line 5F-5F in FIG. 5E.

[0040] FIG. 6A is a schematic view of a light path folding module of the 1st Example according to the 6th Embodiment of the present disclosure.

[0041] FIG. 6B is an exploded view of the light path folding module of the 1st Example according to the 6th Embodiment in FIG. 6A.

[0042] FIG. 6C is a plane view of the light path folding module of the 1st Example according to the 6th Embodiment in FIG. 6A.

[0043] FIG. 6D is a schematic view of the opaque body of the 1st Example according to the 6th Embodiment in FIG. 6A.

[0044] FIG. 6E is another plane view of the light path folding module of the 1st Example according to the 6th Embodiment in FIG. 6A.

[0045] FIG. 6F is a cross-sectional view along line 6F-6F in FIG. 6E.

[0046] FIG. 7A is a schematic view of a light path folding module of the 1st Example according to the 7th Embodiment of the present disclosure.

[0047] FIG. 7B is an exploded view of the light path folding module of the 1st Example according to the 7th Embodiment in FIG. 7A.

[0048] FIG. 7C is a plane view of the light path folding module of the 1st Example according to the 7th Embodiment in FIG. 7A.

[0049] FIG. 7D is a schematic view of the opaque body of the 1st Example according to the 7th Embodiment in FIG. 7A.

[0050] FIG. 7E is another plane view of the light path folding module of the 1st Example according to the 7th Embodiment in FIG. 7A.

[0051] FIG. 7F is a cross-sectional view along line 7F-7F in FIG. 7E.

[0052] FIG. 8A is a schematic view of a light path folding module of the 1st Example according to the 8th Embodiment of the present disclosure.

[0053] FIG. 8B is an exploded view of the light path folding module of the 1st Example according to the 8th Embodiment in FIG. 8A.

[0054] FIG. 8C is a plane view of the light path folding module of the 1st Example according to the 8th Embodiment in FIG. 8A.

[0055] FIG. 8D is a schematic view of the opaque body **820** of the 1st Example according to the 8th Embodiment in FIG. 8A.

[0056] FIG. 8E is another plane view of the light path folding module **800** of the 1st Example according to the 8th Embodiment in FIG. 8A.

[0057] FIG. 8F is a cross-sectional view along line 8F-8F in FIG. 8E.

[0058] FIG. 9A is a schematic view of an electronic device **90** according to the 9th Embodiment of the present disclosure.

[0059] FIG. 9B is another schematic view of the electronic device **90** according to the 9th Embodiment in FIG. 9A.

[0060] FIG. 9C is a schematic view of an image captured by the electronic device **90** according to the 9th Embodiment in FIG. 9A.

[0061] FIG. 9D is a schematic view of an image captured by the electronic device **90** according to the 9th Embodiment in FIG. 9A.

[0062] FIG. 9E is another schematic view of an image captured by the electronic device **90** according to the 9th Embodiment in FIG. 9A.

[0063] FIG. 10 is a schematic view of an electronic device according to the 10th Embodiment of the present disclosure.

[0064] FIG. 11A shows a schematic view of a vehicle device according to the 11th Embodiment of the present disclosure.

[0065] FIG. 11B shows another schematic view of the vehicle device according to the 11th Embodiment in FIG. 11A.

[0066] FIG. 11C shows further another schematic view of the vehicle device according to the 11th Embodiment in FIG. 11A.

DETAILED DESCRIPTION

[0067] The present disclosure provides a light path folding module, which includes a light path folding element and an opaque body, wherein the opaque body is disposed relative to the light path folding element. The light path folding element is for folding an incident optical axis to an exiting optical axis, and the light path folding element includes a light incident surface and a light exiting surface. The incident optical axis passes through and enters into the light incident surface. The exiting optical axis passes through and is away from the light exiting surface. The opaque body includes a non-closed ring structure and a plurality of light blocking structures. The exiting optical axis passes through a notch opening of the non-closed ring structure. The light blocking structures extend towards the exiting optical axis from the non-closed ring structure along a direction perpendicular to the exiting optical axis, wherein the light blocking structures are disposed adjacent to the light exiting surface of the light path folding element. When the exiting optical axis as a center of circle is taken, an angle occupied by the light blocking structures is θ_e , and a number of the light blocking structures is N_e , the following conditions are satisfied: $10^\circ < \theta_e < 350^\circ$; and $15 < N_e < 460$. Therefore, the non-imaging light outside the field of view can be blocked by arranging the light blocking structures towards the direction of the exiting optical axis so as to maintain the image being clear.

[0068] Specifically, the light path folding element can be a prism, a reflector, etc. Each of the light blocking structures can be wedge-shaped, straight strip-shaped, petal-shaped, semi-cylindrical shape or a wavy-shaped, etc. Furthermore, if the direction parallel to the exiting optical axis is Z axis, then the direction perpendicular to the exiting optical axis is any direction on an XY plane defined by X axis and Y axis.

[0069] Furthermore, the non-closed ring structure can be an appearance feature of the opaque body, such as C-shaped, U-shaped, straight-shaped, etc., but the present disclosure is not limited thereto. The light blocking structure can be arranged on the non-closed ring structure substantially along a circumferential direction surrounding the exiting optical axis.

[0070] Each of the light blocking structures can extend towards the exiting optical axis and gradually shrink and intersect. Therefore, the light blocking performance of the light blocking structures can be enhanced.

[0071] The opaque body and the light blocking structures can be formed integrally. Therefore, the assembling process between the opaque body and the light blocking structures can be eliminated to make the manufacturing process faster.

[0072] When the exiting optical axis as the center of circle is taken, the angle occupied by the light blocking structures is θ_e , and a number of the light blocking structures is N_e , the following conditions are satisfied: $10 \text{ degrees} < \theta_e < 150 \text{ degrees}$; and $15 < N_e < 250$. Therefore, the reflection path of non-imaging light on the surface of specific location of the opaque body can be destroyed, so that the non-imaging light cannot enter the imaging lens assembly, thereby improving the image quality. Specifically, the light blocking structures can be arranged at the non-closed ring structure along the straight-shaped.

[0073] When the exiting optical axis as the center of circle is taken, the angle occupied by the light blocking structures is θ_e , and the number of the light blocking structures is N_e , the following conditions are satisfied: $110 \text{ degrees} < \theta_e < 350 \text{ degrees}$; and $50 < N_e < 300$. Therefore, the light blocking requirement of different optical designs can be satisfied so as to improve the optical design margin. Specifically, the light blocking structures can be arranged at the non-closed ring structure along U-shaped.

[0074] Each of the light blocking structures is a wedge-shaped protrusion. When the wedge-shaped protrusion has an included angle θ_e' , the following condition is satisfied: $0 \text{ degrees} < \theta_e' < 90 \text{ degrees}$. Therefore, it is favorable for destroying the reflection path of non-imaging light by the shape design of the wedge-shaped protrusion so as to improve the light blocking effect. Specifically, the opaque body can be a holder or a carrier formed by plastic injection molding, which is transferred into the wedge-shaped or the semi-cylindrical appearance of each of the light blocking structures by the mold. The opaque body can also be a light blocking sheet formed by stamping manufacturing process, which is cut into the wedge-shaped or the petal shaped appearance of each of the light blocking structures via the mold, but the present disclosure is not limited thereto.

[0075] Each of the light blocking structures forms a convex arc. When the convex arc has a curvature radius R_e , the following condition is satisfied: $0 \text{ mm} < R_e < 0.3 \text{ mm}$. Therefore, it is favorable for destroying the reflection path of non-imaging light so as to improve the light blocking effect.

[0076] Each of the light blocking structures forms a concave arc. When the concave arc has a curvature radius R_e' , the following condition is satisfied: $0 \text{ mm} < R_e' < 0.3 \text{ mm}$. Therefore, it is favorable for destroying the reflection path of non-imaging light so as to improve the light blocking effect.

[0077] The non-closed ring structure can further include at least one auxiliary arc, and at least one portion of the light blocking structures is disposed on the auxiliary arc. When the auxiliary arc has a curvature radius r , the following condition is satisfied: $0.5 \text{ mm} < r < 50 \text{ mm}$. Therefore, the flare at the specific angle can be eliminated to make the image being clear. Specifically, each light blocking structure (including the wedge-shaped protrusion, the convex arc, the concave arc or the convex arc and the concave arc arranged continuously) can be further disposed at the auxiliary arc.

[0078] When a height of each of the light blocking structures extending along the direction perpendicular to the exiting optical axis is H_e , the following condition is satisfied: $0.01 \text{ mm} < H_e < 1.2 \text{ mm}$. Therefore, the height extension range with the better light blocking effect is achieved.

[0079] When a length of each of the light blocking structures extending along a direction parallel to the exiting optical axis is L_e , the following condition is satisfied: $0.01 \text{ mm} < L_e < 2.8 \text{ mm}$. Therefore, the length extension range with the better light blocking effect is achieved.

[0080] Furthermore, the light path folding element can be made of plastic material, and the opaque body can be made of black plastic material, but the present disclosure is not limited thereto.

[0081] The present disclosure provides a light path folding module, including a light path folding element and an opaque body, wherein the opaque body is disposed relative to the light path folding

element. The light path folding element is for folding an incident optical axis to an exiting optical axis, and the light path folding element includes a light incident surface and a light exiting surface. The incident optical axis passes through and enters into the light incident surface. The exiting optical axis passes through and is away from the light exiting surface. The opaque body includes a non-closed ring structure and a plurality of light blocking structures. The incident optical axis passes through a notch opening of the non-closed ring structure. The light blocking structures extend toward the incident optical axis from the non-closed ring structure along a direction perpendicular to the incident optical axis, wherein the light blocking structures are disposed adjacent to the light incident surface of the light path folding element. When the incident optical axis as a center of circle, an angle occupied by the light blocking structures is θ_i , and a number of the light blocking structures is N_i , the following conditions are satisfied: $10^\circ < \theta_i < 350^\circ$; and $15 < N_i < 460$. Therefore, the non-imaging light outside the field of view can be blocked by arranging the light blocking structure toward the direction of the incident optical axis so as to maintain the image being clear.

[0082] Each of the light blocking structures can extend toward the incident optical axis and gradually shrink and intersect. Therefore, the light blocking performance of the light blocking structures can be enhanced.

[0083] The opaque body and the light blocking structures can be formed integrally. Therefore, the assembling process between the opaque body and the light blocking structures can be eliminated to make the manufacturing process faster.

[0084] When the incident optical axis as the center of circle is taken, the angle occupied by the light blocking structures is θ_i , and a number of the light blocking structures is N_i , the following conditions are satisfied: $10^\circ < \theta_i < 150^\circ$; and $15 < N_i < 250$. Therefore, the reflection path of non-imaging light on the surface of specific location of the opaque body can be destroyed, so that non-imaging light cannot enter the imaging lens assembly, thereby improving the image quality. Specifically, the light blocking structures can be arranged at the non-closed ring structure along straight-shaped.

[0085] When the incident optical axis as the center of circle is taken, the angle occupied by the light blocking structures is θ_i , and the number of the light blocking structures is N_i , the following conditions are satisfied: $110^\circ < \theta_i < 350^\circ$; and $50 < N_i < 300$. Therefore, the light blocking requirement of different optical designs can be satisfied so as to improve the optical design margin. Specifically, the light blocking structures can be arranged at the non-closed ring structure along U-shaped.

[0086] Each of the light blocking structures is a wedge-shaped protrusion. When the wedge-shaped protrusion has an included angle θ_i' , the following condition is satisfied: $0^\circ < \theta_i' < 90^\circ$. Therefore, it is favorable for destroying the reflection path of non-imaging light by the shape design of the wedge-shaped protrusion so as to improve the light blocking effect.

[0087] Each of the light blocking structures forms a convex arc. When the convex arc has a curvature radius R_i , the following condition is satisfied: $0 \text{ mm} < R_i < 0.3 \text{ mm}$. Therefore, it is favorable for destroying the reflection path of non-imaging light so as to improve the light blocking effect.

[0088] Each of the light blocking structures forms a concave arc. When the concave arc has a curvature radius R_i' , the following condition is satisfied: $0 \text{ mm} < R_i' < 0.3 \text{ mm}$. Therefore, it is favorable for destroying the reflection path of non-imaging light so as to improve the light blocking effect.

[0089] The non-closed ring structure can further include at least one auxiliary arc, and at least one portion of the light blocking structures is disposed on the auxiliary arc. When the auxiliary arc has a curvature radius r , the following condition is satisfied: $0.5 \text{ mm} < r < 50 \text{ mm}$. Therefore, the flare at the specific angle can be eliminated to make the image being clear. Specifically, each light blocking structure (including the wedge-shaped protrusion, the convex arc, the concave arc or the convex arc

and the concave arc arranged continuously) can be further disposed at the auxiliary arc.

[0090] When a height of each of the light blocking structures extending along the direction perpendicular to the incident optical axis is H_i , the following condition is satisfied: $0.01\text{ mm} < H_i < 1.2\text{ mm}$. Therefore, the height extension range with the better light blocking effect is achieved.

[0091] When a length of each of the light blocking structures extending along a direction parallel to the incident optical axis is L_i , the following condition is satisfied: $0.01\text{ mm} < L_i < 2.8\text{ mm}$. Therefore, the length extension range with the better light blocking effect is achieved.

[0092] Furthermore, the light path folding element can be made of plastic material, and the opaque body can be made of black plastic material, but the present disclosure is not limited thereto.

[0093] It must be noted that, in the opaque body of the light path folding module of the present disclosure, the light blocking structures are not limited to being disposed adjacent to one of the light incident surface and the light exiting surface of the light path folding element. The light blocking structures also can be disposed adjacent to the light incident surface and the light exiting surface of the light path folding element at the same time.

[0094] The present disclosure provides a camera module, which includes the light path folding module as described in any of the foregoing aspects and an imaging lens assembly, wherein the imaging lens assembly is disposed adjacent to the light path folding module.

[0095] The present disclosure provides an electronic device, including the camera module as described in any of the foregoing aspects and an image sensor, wherein the image sensor is disposed on an imaging surface of the camera module.

1st Embodiment

[0096] FIG. 1A is a three-dimensional schematic view of a camera module **10** of the 1st Example according to the 1st Embodiment of the present disclosure. FIG. 1B is an exploded view of the camera module **10** of the 1st Example according to the 1st Embodiment in FIG. 1A. FIG. 1C is a cross-sectional view of partial of the camera module **10** of the 1st Example according to the 1st Embodiment in FIG. 1A. In FIG. 1A, FIG. 1B and FIG. 1C, the camera module **10** includes a light path folding module **100** and an imaging lens assembly **11**, wherein the imaging lens assembly **11** is disposed adjacent to the light path folding module **100**. The light path folding module **100** is disposed on an image side of the imaging lens assembly **11** via an assembling component **12**. Specifically, the light path folding module **100** includes a light path folding element **110** and an opaque body **120**, wherein the imaging lens assembly **11** includes a lens barrel portion **11a** for accommodating at least one imaging lens element (not shown in drawings), which is not a technical key point of the present disclosure and will not be described herein.

[0097] FIG. 1D is a plane view of the camera module **10** of the 1st Example according to the 1st Embodiment in FIG. 1A. FIG. 1E is a cross-sectional view along line 1E-1E in FIG. 1D. In FIG. 1A to FIG. 1E, the light path folding element **110** of the light path folding module **100** is disposed relative to the opaque body **120**. The light path folding element **110** is for folding an incident optical axis O_i to an exiting optical axis O_e , which includes a light incident surface **111** and a light exiting surface **112**. The incident optical axis O_i passes through and enters into the light incident surface **111**. The exiting optical axis O_e passes through and is away from the light exiting surface **112**. Specifically, the direction parallel to the exiting optical axis O_e is the Z axis. The direction perpendicular to the exiting optical axis O_e is any direction on an XY plane defined by the X axis and the Y axis. In the 1st Example of the 1st Embodiment, the light path folding element **110** is a prism.

[0098] The opaque body **120** includes a non-closed structure (its reference numeral is omitted) and a plurality of light blocking structures **122**. The incident optical axis O_i passes through a notch opening **121** of the non-closed ring structure. The light blocking structures **122** extend towards the incident optical axis O_i from the non-closed ring structure along a direction perpendicular to the incident optical axis O_i , wherein the light blocking structures **122** are disposed adjacent to the light

incident surface **111** of the light path folding element **110**. Specifically, the light blocking structures **122** are arranged on the non-closed ring structure substantially along a circumferential direction surrounding the incident optical axis O_i . Furthermore, the opaque body **120** and the light blocking structures **122** are formed integrally.

[0099] FIG. **1F** is a plane view of the opaque body **120** of the 1st Example according to the 1st Embodiment in FIG. **1A**. In FIG. **1C** and FIG. **1F**, each of the light blocking structures **122** extends towards the incident optical axis O_i and gradually shrinks and intersects. Each of the light blocking structures **122** is a wedge-shaped protrusion. The non-closed ring structure can further include two auxiliary arcs **123**, and the portion of the light blocking structures **122** is disposed on the auxiliary arcs **123**.

[0100] Furthermore, in the 1st Example of the 1st Embodiment, the light path folding element **110** is made of plastic material. The opaque body **120** is made of black plastic material.

[0101] In FIG. **1E** and FIG. **1F**, according to the 1st Example of the 1st Embodiment, when the incident optical axis O_i as a center of circle is taken, an angle occupied by the light blocking structures **122** is θ_i , a number of the light blocking structures **122** is N_i , the wedge-shaped protrusion has an included angle θ_i' , the auxiliary arc **123** has a curvature radius r , a height of each of the light blocking structures **122** extending along the direction perpendicular to the incident optical axis O_i is H_i , a length of each of the light blocking structures **122** extending along a direction parallel to the incident optical axis O_i is L_i , the following conditions shown in Table 1A are satisfied.

TABLE-US-00001 TABLE 1A 1st Example of 1st Embodiment θ_i (degrees) 318.6 r (mm) 1.50 N_i 141 H_i (mm) 0.25 θ_i' (degrees) 30 L_i (mm) 0.02

[0102] FIG. **1G** is a plane view of the opaque body **120** of the 2nd Example according to the 1st Embodiment in FIG. **1A**. It must be noted that the difference between the 2nd Example of the 1st Embodiment of the present disclosure and the 1st Example thereof is only in the shape of the opaque body **120**, and the other components, positions and connection relationships are the same or similar, which will not be described herein. In FIG. **1C** and FIG. **1G**, according to the 2nd Example of the 1st Embodiment, each of the light blocking structures **122** extends towards the incident optical axis O_i and gradually shrinks and intersects. Each of the light blocking structures **122** is petal-shaped. Each of the light blocking structures **122** forms a concave arc (its reference numeral is omitted). The non-closed ring structure can further include two auxiliary arcs **123**, and the portion of the light blocking structures **122** is disposed on the auxiliary arcs **123**.

[0103] In FIG. **1E** and FIG. **1G**, according to the 2nd Example of the 1st Embodiment, when the incident optical axis O_i as a center of circle is taken, an angle occupied by the light blocking structures **122** is θ_i , a number of the light blocking structures **122** is N_i , the concave arc has a curvature radius R_i' , the auxiliary arc **123** has a curvature radius r , a height of each of the light blocking structures **122** extending along the direction perpendicular to the incident optical axis O_i is H_i , a length of each of the light blocking structures **122** extending along a direction parallel to the incident optical axis O_i is L_i , the following conditions shown in Table 1B are satisfied.

TABLE-US-00002 TABLE 1B 2nd Example of 1st Embodiment θ_i (degrees) 315.3 r (mm) 1.50 N_i 65 H_i (mm) 0.15 R_i' (mm) 0.15 L_i (mm) 0.02

2nd Embodiment

[0104] FIG. **2A** is a three-dimensional view of a camera module **20** of the 1st Example according to the 2nd Embodiment of the present disclosure. FIG. **2B** is an exploded view of the camera module **20** of the 1st Example according to the 2nd Embodiment in FIG. **2A**. In FIG. **2A** and FIG. **2B**, the camera module **20** includes a light path folding module **200** and an imaging lens assembly **21**, wherein the imaging lens assembly **21** is disposed adjacent to the light path folding module **200**. The light path folding module **200** is disposed on an image side of the imaging lens assembly **21** via an assembling component **22**. Specifically, the light path folding module **200** includes a light path folding element **210** and an opaque body **220**, wherein the imaging lens assembly **21** includes

a lens barrel portion **21a** for accommodating at least one imaging lens element (not shown in drawings), which is not a technical key point of the present disclosure and will not be described herein.

[0105] FIG. 2C is a plane view of the camera module **20** of the 1st Example according to the 2nd Embodiment in FIG. 2A. FIG. 2D is a cross-sectional view along line 2D-2D in FIG. 2C. In FIG. 2A to FIG. 2D, the light path folding element **210** of the light path folding module **200** is disposed relative to the opaque body **220**. The light path folding element **210** is for folding an incident optical axis O_i to an exiting optical axis O_e , which includes a light incident surface **211** and a light exiting surface **212**. The incident optical axis O_i passes through and enters into the light incident surface **211**. The exiting optical axis O_e passes through and is away from the light exiting surface **212**. Specifically, in the 1st Example of the 2nd Embodiment of the present disclosure, the light path folding element **210** is a prism, and the light incident surface **211** and the light exiting surface **212** are both located on the same surface of the light path folding element **210**.

[0106] The opaque body **220** includes two non-closed structures (its reference numeral is omitted) and a plurality of light blocking structures **2221**, **2222**. The incident optical axis O_i passes through a notch opening **2211** of the non-closed ring structure. The light blocking structures **2221** extend towards the incident optical axis O_i from the non-closed ring structure along a direction perpendicular to the incident optical axis O_i , wherein the light blocking structures **2221** are disposed adjacent to the light incident surface **211** of the light path folding element **210**. The exiting optical axis O_e passes through a notch opening **2212** of the non-closed ring structure. The light blocking structures **2222** extend towards the exiting optical axis O_e from the non-closed ring structure along a direction perpendicular to the exiting optical axis O_e , wherein the light blocking structures **2222** are disposed adjacent to the light exiting surface **212** of the light path folding element **210**. Furthermore, the opaque body **220** and the light blocking structures **2221**, **2222** are formed integrally.

[0107] FIG. 2E is a plane view of the opaque body **220** of the 1st Example according to the 2nd Embodiment in FIG. 2A. In FIG. 2E, each of the light blocking structures **2221** extends towards the incident optical axis O_i and gradually shrinks and intersects. Each of the light blocking structures **2221** is a wedge-shaped protrusion. Each of the light blocking structures **2222** extends towards the exiting optical axis O_e and gradually shrinks and intersects. Each of the light blocking structures **2222** is a wedge-shaped protrusion.

[0108] Furthermore, in the 1st Example of the 2nd Embodiment, the light path folding element **210** is made of plastic material. The opaque body **220** is made of black plastic material.

[0109] In FIG. 2D and FIG. 2E, according to the 1st Example of the 2nd Embodiment, when the incident optical axis O_i as a center of circle is taken, an angle occupied by the light blocking structures **2221** is θ_i , a number of the light blocking structures **2221** is N_i , the wedge-shaped protrusion of the light blocking structures **2221** has an included angle θ_i' , a height of each of the light blocking structures **2221** extending along the direction perpendicular to the incident optical axis O_i is H_i , a length of each of the light blocking structures **2221** extending along a direction parallel to the incident optical axis O_i is L_i ; when the exiting optical axis O_e as a center of circle is taken, an angle occupied by the light blocking structures **2222** is θ_e , a number of the light blocking structures **2222** is N_e , the wedge-shaped protrusion of the light blocking structures **2222** has an included angle θ_e' , a height of each of the light blocking structures **2222** extending along the direction perpendicular to the exiting optical axis O_e is H_e , a length of each of the light blocking structures **2222** extending along a direction parallel to the exiting optical axis O_e is L_e , the following conditions shown in Table 2A are satisfied.

TABLE-US-00003 TABLE 2A 1st Example of 2nd Embodiment θ_i (degrees) 268.6 θ_e (degrees) 107.7 N_i 74 N_e 26 θ_i' (degrees) 59.5 θ_e' (degrees) 60 H_i (mm) 0.18 H_e (mm) 0.17 L_i (mm) 0.02 L_e (mm) 0.02

[0110] FIG. 2F is a plane view of the opaque body **220** of the 2nd Example according to the 2nd

Embodiment in FIG. 2A. It must be noted that the difference between the 2nd Example of the 2nd Embodiment of the present disclosure and the 1st Example thereof is only in the shape of the opaque body **220**, and the other components, positions and connection relationships are the same or similar, which will not be described herein. In FIG. 2F, according to the 2nd Example of the 2nd Embodiment, each of the light blocking structures **2221** is a petal shaped and forms a convex arc (its reference numeral is omitted). The non-closed ring structure can further include two auxiliary arcs **2231**, and the portion of the light blocking structures **2221** is disposed on the auxiliary arcs **2231**. Each of the light blocking structures **2222** is petal-shaped and forms a concave arc (its reference numeral is omitted).

[0111] In FIG. 2D and FIG. 2F, according to the 2nd Example of the 2nd Embodiment, when the incident optical axis O_i as a center of circle is taken, an angle occupied by the light blocking structures **2221** is θ_i , a number of the light blocking structures **2221** is N_i , the convex arc has a curvature radius R_i , the auxiliary arc **2231** has a curvature radius r , a height of each of the light blocking structures **2221** extending along the direction perpendicular to the incident optical axis O_i is H_i , a length of each of the light blocking structures **2221** extending along a direction parallel to the incident optical axis O_i is L_i ; when the exiting optical axis O_e as a center of circle is taken, an angle occupied by the light blocking structures **2222** is θ_e , a number of the light blocking structures **2222** is N_e , the concave arc has a curvature radius Re' , a height of each of the light blocking structures **2222** extending along the direction perpendicular to the exiting optical axis O_e is He , a length of each of the light blocking structures **2222** extending along a direction parallel to the exiting optical axis O_e is Le , the following conditions shown in Table 2B are satisfied.

TABLE-US-00004 TABLE 2B 2nd Example of 2nd Embodiment θ_i (degrees) 268.2 θ_e (degrees) 104.9 N_i 66 N_e 27 R_i (mm) 0.10 Re' (mm) 0.10 r (mm) 1.61 He (mm) 0.07 H_i (mm) 0.10 Le (mm) 0.02 L_i (mm) 0.02

3rd Embodiment

[0112] FIG. 3A is a schematic view of a light path folding module **300** of the 1st Example according to the 3rd Embodiment of the present disclosure. FIG. 3B is an exploded view of the light path folding module **300** of the 1st Example according to the 3rd Embodiment in FIG. 3A. FIG. 3C is a plane view of the light path folding module **300** of the 1st Example according to the 3rd Embodiment in FIG. 3A. In FIG. 3A, FIG. 3B and FIG. 3C, the light path folding module **300** includes a light path folding element **310** and an opaque body **320**, wherein the opaque body **320** is disposed relative to the light path folding element **310**. Furthermore, the light path folding module **300** can further include a cover **301** and an assembly carrier **302**, wherein the opaque body **320** and the light path folding element **310** are disposed on the assembly carrier **302**. The cover **301** covers the assembly carrier **302**, which is favorable for the overall light path folding module **300** disposing in the camera module or other devices.

[0113] FIG. 3D is a schematic view of the opaque body **320** of the 1st Example according to the 3rd Embodiment in FIG. 3A. FIG. 3E is another plane view of the light path folding module **300** of the 1st Example according to the 3rd Embodiment in FIG. 3A. FIG. 3F is a cross-sectional view along line 3F-3F in FIG. 3E. In FIG. 3D to FIG. 3F, the light path folding element **310** is for folding an incident optical axis O_i to an exiting optical axis O_e , which includes a light incident surface **311** and a light exiting surface **312**. The incident optical axis O_i passes through and enters into the light incident surface **311**. The exiting optical axis O_e passes through and is away from the light exiting surface **312**. In the 1st Example of the 3rd Embodiment, the light path folding element **310** is a prism.

[0114] The opaque body **320** includes a non-closed structure (its reference numeral is omitted) and a plurality of light blocking structures **322**. The exiting optical axis O_e passes through a notch opening **321** of the non-closed ring structure. The light blocking structures **322** extend towards the exiting optical axis O_e from the non-closed ring structure along a direction perpendicular to the exiting optical axis O_e , wherein the light blocking structures **322** are disposed adjacent to the light

exiting surface **312** of the light path folding element **310**. Furthermore, the opaque body **320** and the light blocking structures **322** are formed integrally.

[0115] Each of the light blocking structures **322** extends towards the exiting optical axis O_e and gradually shrinks and intersects. Each of the light blocking structures **322** is a wedge-shaped protrusion.

[0116] Furthermore, in the 1st Example of the 3rd Embodiment, the light path folding element **310** is made of plastic material. The opaque body **320** is made of black plastic material.

[0117] In FIG. 3E and FIG. 3F, according to the 1st Example of the 3rd Embodiment, when the exiting optical axis O_e as a center of circle is taken, an angle occupied by the light blocking structures **322** is θ_e , a number of the light blocking structures **322** is N_e , the wedge-shaped protrusion has an included angle θ_e' , a height of each of the light blocking structures **322** extending along the direction perpendicular to the exiting optical axis O_e is H_e , a length of each of the light blocking structures **322** extending along a direction parallel to the exiting optical axis O_e is L_e , the following conditions shown in Table 3A are satisfied.

TABLE-US-00005 TABLE 3A 1st Example of 3rd Embodiment θ_e (degrees) 129.5 H_e (mm) 0.09 N_e 106 L_e (mm) 0.85 θ_e' (degrees) 60

4th Embodiment

[0118] FIG. 4A is a schematic view of a light path folding module **400** of the 1st Example according to the 4th Embodiment of the present disclosure. FIG. 4B is an exploded view of the light path folding module **400** of the 1st Example according to the 4th Embodiment in FIG. 4A. FIG. 4C is a plane view of the light path folding module **400** of the 1st Example according to the 4th Embodiment in FIG. 4A. In FIG. 4A, FIG. 4B and FIG. 4C, the light path folding module **400** includes a light path folding element **410** and an opaque body **420**, wherein the opaque body **420** is disposed relative to the light path folding element **410**. Furthermore, the light path folding module **400** can further include a cover **401** and an assembly carrier **402**, wherein the opaque body **420** and the light path folding element **410** are disposed on the assembly carrier **402**. The cover **401** covers the assembly carrier **402**, which is favorable for the overall light path folding module **400** disposing in the camera module or other devices.

[0119] FIG. 4D is a schematic view of the opaque body **420** of the 1st Example according to the 4th Embodiment in FIG. 4A. FIG. 4E is another plane view of the light path folding module **400** of the 1st Example according to the 4th Embodiment in FIG. 4A. FIG. 4F is a cross-sectional view along line 4F-4F in FIG. 4E. In FIG. 4D to FIG. 4F, the light path folding element **410** is for folding an incident optical axis O_i to an exiting optical axis O_e , which includes a light incident surface **411** and a light exiting surface **412**. The incident optical axis O_i passes through and enters into the light incident surface **411**. The exiting optical axis O_e passes through and is away from the light exiting surface **412**. In the 1st Example of the 4th Embodiment, the light path folding element **410** is a prism.

[0120] The opaque body **420** includes a non-closed structure (its reference numeral is omitted) and a plurality of light blocking structures **422**. The exiting optical axis O_e passes through a notch opening **421** of the non-closed ring structure. The light blocking structures **422** extend towards the exiting optical axis O_e from the non-closed ring structure along a direction perpendicular to the exiting optical axis O_e , wherein the light blocking structures **422** are disposed adjacent to the light exiting surface **412** of the light path folding element **410**. Furthermore, the opaque body **420** and the light blocking structures **422** are formed integrally.

[0121] Each of the light blocking structures **422** extends towards the exiting optical axis O_e and gradually shrinks and intersects. Each of the light blocking structures **422** is a wedge-shaped protrusion. Specifically, the light blocking structures **422** is arranged on the non-closed ring structure substantially along a circumferential direction surrounding the exiting optical axis O_e . Furthermore, the light blocking structures **422** are generally arranged in a U-shaped distribution with the exiting optical axis O_e as a center. In the U-shaped distribution, the shape and the size of

the light blocking structures **422** located at the bottom of the U-shaped and the light blocking structures **422** located at the both sides of the U-shaped are slightly different (the differences are shown in Table 4A as follows).

[0122] Furthermore, in the 1st Example of the 4th Embodiment, the light path folding element **410** is made of plastic material. The opaque body **420** is made of black plastic material.

[0123] In FIG. 4E and FIG. 4F, according to the 1st Example of the 4th Embodiment, when the exiting optical axis O_e as a center of circle is taken, an angle occupied by the light blocking structures **422** is θ_e , a number of the light blocking structures **422** is N_e , the wedge-shaped protrusion has an included angle θ_e' , a height of each of the light blocking structures **422** extending along the direction perpendicular to the exiting optical axis O_e is H_e , a length of each of the light blocking structures **422** extending along a direction parallel to the exiting optical axis O_e is L_e , the following conditions shown in Table 4A are satisfied. It must be noted that, as shown in Table 4A, in the columns of θ_e' and H_e , the value on the left side is the value of the light blocking structures **422** located at the both sides of the U-shaped, and the value on the right side is the value of the light blocking structures **422** located at the bottom of the U-shaped.

TABLE-US-00006 TABLE 4A 1st Example of 4th Embodiment θ_e (degrees) 219.8 H_e (mm) 0.12, 0.09 N_e 194 L_e (mm) 0.87 θ_e' (degrees) 45, 60

5th Embodiment

[0124] FIG. 5A is a schematic view of a light path folding module **500** of the 1st Example according to the 5th Embodiment of the present disclosure. FIG. 5B is an exploded view of the light path folding module **500** of the 1st Example according to the 5th Embodiment in FIG. 5A. FIG. 5C is a plane view of the light path folding module **500** of the 1st Example according to the 5th Embodiment in FIG. 5A. In FIG. 5A, FIG. 5B and FIG. 5C, the light path folding module **500** includes a light path folding element **510** and an opaque body **520**, wherein the opaque body **520** is disposed relative to the light path folding element **510**. Furthermore, the light path folding module **500** can further include a cover **501** and an assembly carrier **502**, wherein the opaque body **520** and the light path folding element **510** are disposed on the assembly carrier **502**. The cover **501** covers the assembly carrier **502**, which is favorable for the overall light path folding module **500** disposing in the camera module or other devices.

[0125] FIG. 5D is a schematic view of the opaque body **520** of the 1st Example according to the 5th Embodiment in FIG. 5A. FIG. 5E is another plane view of the light path folding module **500** of the 1st Example according to the 5th Embodiment in FIG. 5A. FIG. 5F is a cross-sectional view along line 5F-5F in FIG. 5E. In FIG. 5D to FIG. 5F, the light path folding element **510** is for folding an incident optical axis O_i to an exiting optical axis O_e , which includes a light incident surface **511** and a light exiting surface **512**. The incident optical axis O_i passes through and enters into the light incident surface **511**. The exiting optical axis O_e passes through and is away from the light exiting surface **512**. In the 1st Example of the 5th Embodiment, the light path folding element **510** is a prism.

[0126] The opaque body **520** includes a non-closed structure (its reference numeral is omitted) and a plurality of light blocking structures **522**. The exiting optical axis O_e passes through a notch opening **521** of the non-closed ring structure. The light blocking structures **522** extend toward the exiting optical axis θ_e from the non-closed ring structure along a direction perpendicular to the exiting optical axis O_e , wherein the light blocking structures **522** are disposed adjacent to the light exiting surface **512** of the light path folding element **510**.

[0127] Furthermore, the opaque body **520** and the light blocking structures **522** are formed integrally.

[0128] The light blocking structures **522** are the wavy shaped. Specifically, a part of the light blocking structures **522** forms a convex arc (its reference numeral is omitted), and a part of the light blocking structures **522** forms a concave arc (its reference numeral is omitted). The convex arc and the concave arc are arranged alternately.

[0129] Furthermore, in the 1st Example of the 5th Embodiment, the light path folding element **510** is made of plastic material. The opaque body **520** is made of black plastic material.

[0130] In FIG. 5E and FIG. 5F, according to the 1st Example of the 5th Embodiment, when the exiting optical axis O_e as a center of circle is taken, an angle occupied by the light blocking structures **522** is θ_e , a number of the light blocking structures **522** is N_e , the convex arc has a curvature radius R_e , the concave arc has a curvature radius R_e' , a height of each of the light blocking structures **522** extending along the direction perpendicular to the exiting optical axis O_e is H_e , a length of each of the light blocking structures **522** extending along a direction parallel to the exiting optical axis O_e is L_e , the following conditions shown in Table 5A are satisfied.

TABLE-US-00007 TABLE 5A 1st Example of 5th Embodiment θ_e (degrees) 134.2 H_e (mm) 0.30 N_e 105 L_e (mm) 0.92 R_e (mm) 0.05 R_e' (mm) 0.05

6th Embodiment

[0131] FIG. 6A is a schematic view of a light path folding module **600** of the 1st Example according to the 6th Embodiment of the present disclosure. FIG. 6B is an exploded view of the light path folding module **600** of the 1st Example according to the 6th Embodiment in FIG. 6A. FIG. 6C is a plane view of the light path folding module **600** of the 1st Example according to the 6th Embodiment in FIG. 6A. In FIG. 6A, FIG. 6B and FIG. 6C, the light path folding module **600** includes a light path folding element **610** and an opaque body **620**, wherein the opaque body **620** is disposed relative to the light path folding element **610**. Furthermore, the light path folding module **600** can further include a cover **601** and an assembly carrier **602**, wherein the opaque body **620** and the light path folding element **610** are disposed on the assembly carrier **602**. The cover **601** covers the assembly carrier **602**, which is favorable for the overall light path folding module **600** disposing in the camera module or other devices.

[0132] FIG. 6D is a schematic view of the opaque body **620** of the 1st Example according to the 6th Embodiment in FIG. 6A. FIG. 6E is another plane view of the light path folding module **600** of the 1st Example according to the 6th Embodiment in FIG. 6A. FIG. 6F is a cross-sectional view along line 6F-6F in FIG. 6E. In FIG. 6D to FIG. 6F, the light path folding element **610** is for folding an incident optical axis O_i to an exiting optical axis O_e , which includes a light incident surface **611** and a light exiting surface **612**. The incident optical axis O_i passes through and enters into the light incident surface **611**. The exiting optical axis O_e passes through and is away from the light exiting surface **612**. In the 1st Example of the 6th Embodiment, the light path folding element **610** is a prism.

[0133] The opaque body **620** includes a non-closed structure (its reference numeral is omitted) and a plurality of light blocking structures **622**. The exiting optical axis O_e passes through a notch opening **621** of the non-closed ring structure. The light blocking structures **622** extend towards the exiting optical axis O_e from the non-closed ring structure along a direction perpendicular to the exiting optical axis O_e , wherein the light blocking structures **622** are disposed adjacent to the light exiting surface **612** of the light path folding element **610**. Furthermore, the opaque body **620** and the light blocking structures **622** are formed integrally.

[0134] Each of the light blocking structures **622** extends toward the exiting optical axis O_e and gradually shrinks and intersects. Each of the light blocking structures **622** is a wedge-shaped protrusion. Specifically, the light blocking structures **622** are arranged on the non-closed ring structure substantially along a circumferential direction surrounding the exiting optical axis O_e . Furthermore, the light blocking structures **622** are generally arranged in a U-shaped distribution with the exiting optical axis O_e as a center. In the U-shaped distribution, the shape and the size of the light blocking structures **622** located at the bottom of the U-shaped and the light blocking structures **622** located at the both sides of the U-shaped are slightly different (the differences are shown in Table 6A as follows).

[0135] Furthermore, the non-closed ring structure can further include a plurality of auxiliary arcs **623**, and the light blocking structures **622** are disposed on the auxiliary arcs **623**. Specifically, the

shape and the size of the auxiliary arcs **623** located at the bottom of the U-shaped and the auxiliary arcs **623** located at the both sides of the U-shaped are slightly different (the differences are shown in Table 6A as follows).

[0136] Furthermore, in the 1st Example of the 6th Embodiment, the light path folding element **610** is made of plastic material. The opaque body **620** is made of black plastic material.

[0137] In FIG. 6E and FIG. 6F, according to the 1st Example of the 6th Embodiment, when the exiting optical axis O_e as a center of circle is taken, an angle occupied by the light blocking structures **622** is θ_e , a number of the light blocking structures **622** is N_e , the wedge-shaped protrusion has an included angle θ_e' , the auxiliary arcs **623** has a curvature radius r , a height of each of the light blocking structures **622** extending along the direction perpendicular to the exiting optical axis O_e is H_e , a length of each of the light blocking structures **622** extending along a direction parallel to the exiting optical axis O_e is L_e , the following conditions shown in Table 6A are satisfied. It must be noted that, as shown in Table 6A, in the columns of θ_e' , r and H_e , the value on the left side is the value of the light blocking structures **622** and the auxiliary arcs **623** located at the bottom of the U-shaped, and the value on the right side is the value of the light blocking structures **622** and the auxiliary arcs **623** located at the both sides of the U-shaped.

TABLE-US-00008 TABLE 6A 1st Example of 6th Embodiment θ_e (degrees) 219.2 H_e (mm) 0.06, 0.07 N_e 232 L_e (mm) 0.87 θ_e' (degrees) 55.6, 78.8 r (mm) 9.0, 27.4

7th Embodiment

[0138] FIG. 7A is a schematic view of a light path folding module **700** of the 1st Example according to the 7th Embodiment of the present disclosure. FIG. 7B is an exploded view of the light path folding module **700** of the 1st Example according to the 7th Embodiment in FIG. 7A. FIG. 7C is a plane view of the light path folding module **700** of the 1st Example according to the 7th Embodiment in FIG. 7A. In FIG. 7A, FIG. 7B and FIG. 7C, the light path folding module **700** includes a light path folding element **710** and an opaque body **720**, wherein the opaque body **720** is disposed relative to the light path folding element **710**. Furthermore, the light path folding module **700** can further include a cover **701** and an assembly carrier **702**, wherein the opaque body **720** and the light path folding element **710** are disposed on the assembly carrier **702**. The cover **701** covers the assembly carrier **702**, which is favorable for the overall light path folding module **700** disposing in the camera module or other devices.

[0139] FIG. 7D is a schematic view of the opaque body **720** of the 1st Example according to the 7th Embodiment in FIG. 7A. FIG. 7E is another plane view of the light path folding module **700** of the 1st Example according to the 7th Embodiment in FIG. 7A. FIG. 7F is a cross-sectional view along line 7F-7F in FIG. 7E. In FIG. 7D to FIG. 7F, the light path folding element **710** is for folding an incident optical axis O_i to an exiting optical axis O_e , which includes a light incident surface **711** and a light exiting surface **712**. The incident optical axis O_i passes through and enters into the light incident surface **711**. The exiting optical axis O_e passes through and is away from the light exiting surface **712**. In the 1st Example of the 7th Embodiment, the light path folding element **710** is a prism.

[0140] The opaque body **720** includes a non-closed structure (its reference numeral is omitted) and a plurality of light blocking structures **722**. The incident optical axis O_i passes through a notch opening **721** of the non-closed ring structure. The light blocking structures **722** extend towards the incident optical axis O_i from the non-closed ring structure along a direction perpendicular to the incident optical axis O_i , wherein the light blocking structures **722** are disposed adjacent to the light incident surface **711** of the light path folding element **710**. Furthermore, the opaque body **720** and the light blocking structures **722** are formed integrally.

[0141] Each of the light blocking structures **722** extends toward the incident optical axis O_i and gradually shrinks and intersects. Each of the light blocking structures **722** is a wedge-shaped protrusion.

[0142] Furthermore, in the 1st Example of the 7th Embodiment, the light path folding element **710**

is made of plastic material. The opaque body **720** is made of black plastic material.

[0143] In FIG. 7C and FIG. 7F, according to the 1st Example of the 7th Embodiment, when the incident optical axis O_i as a center of circle is taken, an angle occupied by the light blocking structures **722** is θ_i , a number of the light blocking structures **722** is N_i , the wedge-shaped protrusion has an included angle θ_i' , a height of each of the light blocking structures **722** extending along the direction perpendicular to the incident optical axis O_i is H_i , a length of each of the light blocking structures **722** extending along a direction parallel to the incident optical axis O_i is L_i , the following conditions shown in Table 7A are satisfied.

TABLE-US-00009 TABLE 7A 1st Example of 7th Embodiment θ_i (degrees) 123.6 H_i (mm) 0.35 N_i 220 L_i (mm) 0.35 θ_i' (degrees) 20

8th Embodiment

[0144] FIG. 8A is a schematic view of a light path folding module **800** of the 1st Example according to the 8th Embodiment of the present disclosure. FIG. 8B is an exploded view of the light path folding module **800** of the 1st Example according to the 8th Embodiment in FIG. 8A. FIG. 8C is a plane view of the light path folding module **800** of the 1st Example according to the 8th Embodiment in FIG. 8A. In FIG. 8A, FIG. 8B and FIG. 8C, the light path folding module **800** includes a light path folding element **810** and an opaque body **820**, wherein the opaque body **820** is disposed relative to the light path folding element **810**. Furthermore, the light path folding module **800** can further include a cover **801** and an assembly carrier **802**, wherein the opaque body **820** and the light path folding element **810** are disposed on the assembly carrier **802**. The cover **801** covers the assembly carrier **802**, which is favorable for the overall light path folding module **800** disposing in the camera module or other devices.

[0145] FIG. 8D is a schematic view of the opaque body **820** of the 1st Example according to the 8th Embodiment in FIG. 8A. FIG. 8E is another plane view of the light path folding module **800** of the 1st Example according to the 8th Embodiment in FIG. 8A. FIG. 8F is a cross-sectional view along line 8F-8F in FIG. 8E. In FIG. 8D to FIG. 8F, the light path folding element **810** is for folding an incident optical axis O_i to an exiting optical axis O_e , which includes a light incident surface **811** and a light exiting surface **812**. The incident optical axis O_i passes through and enters into the light incident surface **811**. The exiting optical axis O_e passes through and is away from the light exiting surface **812**. In the 1st Example of the 8th Embodiment, the light path folding element **810** is a prism.

[0146] The opaque body **820** includes a non-closed structure (its reference numeral is omitted) and a plurality of light blocking structures **822**. The exiting optical axis O_e passes through a notch opening **821** of the non-closed ring structure. The light blocking structures **822** extend towards the exiting optical axis O_e from the non-closed ring structure along a direction perpendicular to the exiting optical axis O_e , wherein the light blocking structures **822** are disposed adjacent to the light exiting surface **812** of the light path folding element **810**. Furthermore, the opaque body **820** and the light blocking structures **822** are formed integrally.

[0147] The light blocking structures **822** are wavy-shaped. Specifically, a part of the light blocking structures **822** forms a convex arc (its reference numeral is omitted), and a part of the light blocking structures **822** forms a concave arc (its reference numeral is omitted). The convex arc and the concave arc are arranged alternately. Furthermore, the non-closed ring structure can further include a plurality of auxiliary arcs **823**, and the light blocking structures **822** are disposed on the auxiliary arcs **823**.

[0148] Furthermore, in the 1st Example of the 8th Embodiment, the light path folding element **810** is made of plastic material. The opaque body **820** is made of black plastic material.

[0149] In FIG. 8C, FIG. 8E and FIG. 8F, according to the 1st Example of the 8th Embodiment, when the exiting optical axis O_e as a center of circle is taken, an angle occupied by the light blocking structures **822** is θ_e , a number of the light blocking structures **822** is N_e , the convex arc has a curvature radius R_e , the concave arc has a curvature radius R_e' , the auxiliary arcs **823** has a

curvature radius r , a height of each of the light blocking structures **822** extending along the direction perpendicular to the exiting optical axis O_e is H_e , a length of each of the light blocking structures **822** extending along a direction parallel to the exiting optical axis O_e is L_e , the following conditions shown in Table 8A are satisfied.

TABLE-US-00010 TABLE 8A 1st Example of 8th Embodiment θ_e (degrees) 130.6 H_e (mm) 0.59
 N_e 105 L_e (mm) 1.40 R_e (mm) 0.05 R_e' (mm) 0.05 r (mm) 2

9th Embodiment

[0150] FIG. **9A** is a schematic view of an electronic device **90** according to the 9th Embodiment of the present disclosure. FIG. **9B** is another schematic view of the electronic device **90** according to the 9th Embodiment in FIG. **9A**. In FIG. **9A** and FIG. **9B**, the electronic device **90** is a smart phone, and includes a plurality of camera modules, a plurality of image sensors and a user interface **97**, wherein the image sensors are disposed on an imaging surface of each camera module, respectively. Furthermore, the camera module includes a high-pixel camera module **91**, an ultra-wide angle camera module **92** and two telephoto camera modules **93**, **94**, and the user interface **97** is a touch screen, which is not limited thereto. Specifically, each camera module can include the light path folding module according to any one example of the aforementioned 1st Embodiment to 8th Embodiment and an imaging lens assembly, and the imaging lens assembly is disposed adjacent to the light path folding module, but the present disclosure is not limited thereto.

[0151] Furthermore, users enter a shooting mode via the user interface **97**, wherein the user interface **97** is for displaying the scene, and the shooting angle can be manually adjusted to switch the different camera modules. At this moment, the imaging light is gathered on the image sensor via the camera module, and an electronic signal about an image is output to an image signal processor (ISP) **95**.

[0152] In FIG. **9A**, to meet a specification of the electronic device **90**, the electronic device **90** can further include an optical anti-shake mechanism (not shown in drawings). Furthermore, the electronic device **90** can further include at least one focusing assisting module (not shown in drawings) and at least one sensing element (not shown in drawings). The focusing assisting module can be a flash module **96** for compensating a color temperature, an infrared distance measurement component, a laser focus module, etc. The sensing element can have functions for sensing physical momentum and kinetic energy, such as an accelerator, a gyroscope, a Hall Effect Element, to sense shaking or jitters applied by hands of the user or external environments. Accordingly, the camera module of the electronic device **90** equipped with an auto-focusing mechanism and the optical anti-shake mechanism can be enhanced to achieve the superior image quality. Furthermore, the electronic device **90** according to the present disclosure can have a capturing function with multiple modes, such as taking optimized selfies, high dynamic range (HDR) under a low light condition, 4K resolution recording, etc. Furthermore, the users can visually see a captured image of the camera through the user interface **97** and manually operate the view finding range on the user interface **97** to achieve the auto-focus function of what you see is what you get.

[0153] Moreover, the camera module, the optical anti-shake mechanism, the sensing element and the focusing assisting module can be disposed on a flexible printed circuit board (FPC) (not shown in drawings) and electrically connected to the associated components, such as the image signal processor **95**, via a connector (not shown in drawings) to perform a capturing process. Since the current electronic devices, such as smart phones, have a tendency of being compact, the way of firstly disposing the camera module and related components on the flexible printed circuit board and secondly integrating the circuit thereof into the main board of the electronic device via the connector can satisfy the requirements of the mechanical design and the circuit layout of the limited space inside the electronic device, and obtain more margins. The autofocus function of the camera module can also be controlled more flexibly via the touch screen of the electronic device. According to the 9th Embodiment, the electronic device **90** can include a plurality of sensing elements and a plurality of focusing assisting modules. The sensing elements and the focusing

assisting modules are disposed on the flexible printed circuit board and at least one other flexible printed circuit board (not shown in drawings) and electrically connected to the associated components, such as the image signal processor **95**, via corresponding connectors to perform the capturing process. In other examples (not shown in drawings), the sensing elements and the focusing assisting modules can also be disposed on the main board of the electronic device or carrier boards of other types according to requirements of the mechanical design and the circuit layout.

[0154] Furthermore, the electronic device **90** can further include, but not be limited to, a display, a control unit, a storage unit, a random access memory (RAM), a read-only memory (ROM), or the combination thereof.

[0155] FIG. **9C** is a schematic view of an image captured by the electronic device **90** according to the 9th Embodiment in FIG. **9A**. In FIG. **9C**, the larger range of the image can be captured via the ultra-wide angle camera module **92**, and the ultra-wide angle camera module **92** can have the function of accommodating more wide range of the scene.

[0156] FIG. **9D** is another schematic view of an image captured by the electronic device **90** according to the 9th Embodiment in FIG. **9A**. In FIG. **9D**, the image of the certain range with the high resolution can be captured via the high resolution camera module **91**, and the high resolution camera module **91** has the function of the high resolution and the low deformation.

[0157] FIG. **9E** is further another schematic view of an image captured by the electronic device **90** according to the 9th Embodiment in FIG. **9A**. In FIG. **9E**, the telephoto camera modules **93**, **94** have the enlarging function of the high magnification, and the distant image can be captured and enlarged with high magnification via the telephoto camera modules **93**, **94**.

[0158] In FIG. **9C** to FIG. **9E**, the zooming function can be obtained via the electronic device **90**, when the scene is captured via the camera module with different focal lengths cooperated with the function of image processing.

10th Embodiment

[0159] FIG. **10** is a schematic view of an electronic device **90a** according to the 10th Embodiment of the present disclosure. In FIG. **10**, the electronic device **90a** is a smart phone, and includes a plurality of camera modules and a plurality of image sensors, wherein the image sensors are disposed on an imaging surface of each camera module, respectively. Furthermore, the camera module includes the ultra-wide angle camera modules **91a**, **92a**, the wide angle camera modules **93a**, **94a**, the telephoto camera modules **95a**, **96a**, **97a**, **98a**, and a Time-Of-Flight (TOF) module **99a**. The TOF module **99a** can be another type of the camera module, and the disposition is not limited thereto. In detail, the camera modules can include the light path folding module according to any one example of the aforementioned 1st Embodiment to 8th Embodiment and an imaging lens assembly, and the imaging lens assembly is disposed adjacent to the light path folding module, but the present disclosure is not limited thereto.

[0160] Further, the telephoto camera modules **97a**, **98a** are configured to fold the light path, but the present disclosure is not limited thereto.

[0161] To meet a specification of the electronic device **90a**, the electronic device **90a** can further include an optical anti-shake mechanism (not shown in drawings). Furthermore, the electronic device **90a** can further include at least one focusing assisting module (not shown in drawings) and at least one sensing element (not shown in drawings). The focusing assisting module can be a flash module **901a** for compensating a color temperature, an infrared distance measurement component, a laser focus module, etc. The sensing element can have functions for sensing physical momentum and kinetic energy, such as an accelerator, a gyroscope, a Hall Effect Element, to sense shaking or jitters applied by hands of the user or external environments. Accordingly, the camera module of the electronic device **90a** equipped with an auto-focusing mechanism and the optical anti-shake mechanism can be enhanced to achieve the superior image quality. Furthermore, the electronic device **90a** according to the present disclosure can have a capturing function with multiple modes,

such as taking optimized selfies, High Dynamic Range (HDR) under a low light condition, 4K Resolution recording, etc.

[0162] Further, all of other structures and dispositions according to the 10th Embodiment are the same as the structures and the dispositions according to the 9th Embodiment, and will not be described again herein.

11th Embodiment

[0163] FIG. 11A is a schematic view of a vehicle device **90b** according to the 11th Embodiment of the present disclosure. FIG. 11B is another schematic view of the vehicle device **90b** according to the 11th Embodiment in FIG. 11A. FIG. 11C is further another schematic view of the vehicle device **90b** according to the 11th Embodiment in FIG. 11A. In FIG. 11A to FIG. 11C, the vehicle device **90b** includes a plurality of camera modules **91b** and a plurality of image sensors, wherein the image sensors are disposed on an imaging surface of each camera module **91b**, respectively. In the 11th Embodiment, a number of the camera modules **91b** is six, and the camera modules **91b** can include the light path folding module according to any one example of the aforementioned 1st Embodiment to 8th Embodiment and an imaging lens assembly, and the imaging lens assembly is disposed adjacent to the light path folding module, but the present disclosure is not limited thereto.

[0164] In FIG. 11A to FIG. 11B, the camera module **91b** is a vehicle camera module, and two of the camera modules **91b** are located under two rear view mirrors on the left side and the right side, respectively. Each of the two camera modules **91b** captures image information from a field of view A. Specifically, the field of view A can satisfy the following condition: $40^\circ < A < 90^\circ$. Hence, the image information in the region of two lanes on the left side and the right side can be captured.

[0165] In FIG. 11B, another two of the camera modules **91b** can be disposed in an inner space of the vehicle device **90b**. Specifically, the aforementioned two camera modules **91b** can be disposed near a rear view mirror in the vehicle device **90b** and near a rear window, respectively, or can be disposed on non-mirror surfaces of two rear view mirrors on left side and right side of the vehicle device **90b**, respectively, but not limited thereto.

[0166] In FIG. 11C, further another two of the camera modules **91b** can be disposed at the front end and the rear end of the vehicle device **90b**. Furthermore, the traffic information outside the vehicle can be identified helpfully by the arrangement of the camera modules **91b** disposed at the front end, the rear end and below the left and right rear view mirrors of the vehicle device **90b**. The traffic information outside the vehicle can be 11, 12, 13, 14, but not limited thereto. Therefore, the angle of view can be provided widely to decrease the blind spot, which is favorable for improving driving safety. Furthermore, it is favorable for identifying the external space information out of the vehicle device **90b** by arranging the camera modules **91b** around the vehicle device **90b** to achieve the function of autopilot.

[0167] The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

Claims

1. A light path folding module, comprising: a light path folding element for folding an incident optical axis to an exiting optical axis, and the light path folding element comprising: a light incident surface, wherein the incident optical axis passes through and enters into the light incident

surface; and a light exiting surface, wherein the exiting optical axis passes through and is away from the light exiting surface; and an opaque body disposed relative to the light path folding element, and the opaque body comprising: a non-closed ring structure, wherein the exiting optical axis passes through a notch opening of the non-closed ring structure; and a plurality of light blocking structures extending towards the exiting optical axis from the non-closed ring structure along a direction perpendicular to the exiting optical axis, wherein the plurality of light blocking structures are disposed adjacent to the light exiting surface of the light path folding element; wherein when the exiting optical axis as a center of circle is taken, an angle occupied by the plurality of light blocking structures is θ_e , a number of the plurality of light blocking structures is N_e , and the following conditions are satisfied: $10\text{degrees} < \theta_e < 350\text{degrees}$; and $15 < N_e < 460$.

2. The light path folding module of claim 1, wherein each of the plurality of light blocking structures extends towards the exiting optical axis and gradually shrinks and intersects.
3. The light path folding module of claim 1, wherein the opaque body and the plurality of light blocking structures are formed integrally.
4. The light path folding module of claim 1, wherein when the exiting optical axis as the center of circle is taken, the angle occupied by the plurality of light blocking structures is θ_e , the number of the plurality of light blocking structures is N_e , and the following conditions are satisfied: $10\text{degrees} < \theta_e < 150\text{degrees}$; and $15 < N_e < 250$.
5. The light path folding module of claim 1, wherein when the exiting optical axis as the center of circle is taken, the angle occupied by the plurality of light blocking structures is θ_e , the number of the plurality of light blocking structures is N_e , and the following conditions are satisfied: $110\text{degrees} < \theta_e < 350\text{degrees}$; and $50 < N_e < 300$.
6. The light path folding module of claim 1, wherein each of the plurality of light blocking structures is a wedge-shaped protrusion, the wedge-shaped protrusion has an included angle $\theta_{e'}$, and the following condition is satisfied: $0\text{degrees} < \theta_{e'} < 90\text{degrees}$.
7. The light path folding module of claim 1, wherein each of the plurality of light blocking structures forms a convex arc, the convex arc has a curvature radius R_e , and the following condition is satisfied: $0\text{mm} < R_e < 0.3\text{mm}$.
8. The light path folding module of claim 1, wherein each of the plurality of light blocking structures forms a concave arc, the concave arc has a curvature radius $R_{e'}$, and the following condition is satisfied: $0\text{mm} < R_{e'} < 0.3\text{mm}$.
9. The light path folding module of claim 1, wherein the non-closed ring structure further comprises at least one auxiliary arc, at least one portion of the plurality of light blocking structures is disposed on the at least one auxiliary arc, the at least one auxiliary arc has a curvature radius r , and the following condition is satisfied: $0.5\text{mm} < r < 50\text{mm}$.
10. The light path folding module of claim 1, wherein a height of each of the plurality of light blocking structures extends along the direction perpendicular to the exiting optical axis is H_e , and the following condition is satisfied: $0.1\text{mm} < H_e < 1.2\text{mm}$.
11. The light path folding module of claim 1, wherein a length of each of the plurality of light blocking structures extending along a direction parallel to the exiting optical axis is L_e , and the following condition is satisfied: $0.01\text{mm} < L_e < 2.8\text{mm}$.
12. The light path folding module of claim 1, wherein the light path folding element is made of plastic material.
13. The light path folding module of claim 1, wherein the opaque body is made of black plastic material.
14. A camera module, comprising: the light path folding module of claim 1; and an imaging lens assembly disposed adjacent to the light path folding module.
15. An electronic device, comprising: the camera module of claim 14; and an image sensor disposed on an imaging surface of the camera module.
16. A light path folding module, comprising: a light path folding element for folding an incident

optical axis to an exiting optical axis, and the light path folding element comprising: a light incident surface, wherein the incident optical axis passes through and enters into the light incident surface; and a light exiting surface, wherein the exiting optical axis passes through and is away from the light exiting surface; and an opaque body disposed relative to the light path folding element, and the opaque body comprising: a non-closed ring structure, wherein the incident optical axis passes through a notch opening of the non-closed ring structure; and a plurality of light blocking structures extending towards the incident optical axis from the non-closed ring structure along a direction perpendicular to the incident optical axis, wherein the plurality of light blocking structures are disposed adjacent to the light incident surface of the light path folding element; wherein when the incident optical axis as a center of circle is taken, an angle occupied by the plurality of light blocking structures is θ_i , a number of the plurality of light blocking structures is N_i , and the following conditions are satisfied: $10\text{degrees} < \theta_i < 350\text{degrees}$; and $15 < N_i < 460$.

17. The light path folding module of claim 16, wherein each of the plurality of light blocking structures extends towards the incident optical axis and gradually shrinks and intersects.

18. The light path folding module of claim 16, wherein the opaque body and the plurality of light blocking structures are formed integrally.

19. The light path folding module of claim 16, wherein when the incident optical axis as the center of circle is taken, the angle occupied by the plurality of light blocking structures is θ_i , the number of the plurality of light blocking structures is N_i , and the following conditions are satisfied: $10\text{degrees} < \theta_i < 150\text{degrees}$; and $15 < N_i < 250$.

20. The light path folding module of claim 16, wherein when the incident optical axis as the center of circle is taken, the angle occupied by the plurality of light blocking structures is θ_i , the number of the plurality of light blocking structures is N_i , and the following conditions are satisfied: $10\text{degrees} < \theta_i < 350\text{degrees}$; and $50 < N_i < 300$.

21. The light path folding module of claim 16, wherein each of the plurality of light blocking structures is a wedge-shaped protrusion, the wedge-shaped protrusion has an included angle θ_i' , and the following condition is satisfied: $0\text{degrees} < \theta_i' < 90\text{degrees}$.

22. The light path folding module of claim 16, wherein each of the plurality of light blocking structures forms a convex arc, the convex arc has a curvature radius R_i , and the following condition is satisfied: $0\text{mm} < R_i < 0.3\text{mm}$.

23. The light path folding module of claim 16, wherein each of the plurality of light blocking structures forms a concave arc, the concave arc has a curvature radius R_i' , and the following condition is satisfied: $0\text{mm} < R_i' < 0.3\text{mm}$.

24. The light path folding module of claim 16, wherein the non-closed ring structure further comprises at least one auxiliary arc, at least one portion of the plurality of light blocking structures is disposed on the at least one auxiliary arc, the at least one auxiliary arc has a curvature radius r , and the following condition is satisfied: $0.5\text{mm} < r < 50\text{mm}$.

25. The light path folding module of claim 16, wherein a height of each of the plurality of light blocking structures extending along the direction perpendicular to the incident optical axis is H_i , and the following condition is satisfied: $0.01\text{mm} < H_i < 1.2\text{mm}$.

26. The light path folding module of claim 16, wherein a length of each of the plurality of light blocking structures extending along a direction parallel to the incident optical axis is L_i , and the following condition is satisfied: $0.1\text{mm} < L_i < 2.8\text{mm}$.

27. The light path folding module of claim 16, wherein the light path folding element is made of plastic material.

28. The light path folding module of claim 16, wherein the opaque body is made of black plastic material.
